

Extremal Least Common Multiples in Rows of Pascal's Triangle

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Abstract

For $r \geq 0$, let

$$\mathcal{P}_r = \left\{ \binom{r}{0}, \binom{r}{1}, \dots, \binom{r}{\lfloor r/2 \rfloor} \right\}$$

be the set of distinct entries in row r of Pascal's triangle. We study the least possible least common multiple of n entries chosen from one row, with the row itself also free:

$$a(n) = \min_{\substack{r \geq 0, \\ S \subseteq \mathcal{P}_r \\ |S|=n}} \text{lcm}(S).$$

We first recast the fixed-row problem exactly as a weighted prime-power exclusion problem. This structural description explains why optimal supports may develop holes and yields an exact certification method for finite cases. Our main asymptotic result is

$$\log a(n) = 2n + O\left(n \exp\left[-c \frac{(\log n)^{3/5}}{(\log \log n)^{1/5}}\right]\right)$$

for some absolute $c > 0$, so $a(n)^{1/n} \rightarrow e^2$. A two-band refinement further shows that every optimal row satisfies

$$r_n = 2n + O\left(ne^{-c\Phi(n)}\right),$$

and that the minimum prefix defect of an optimal support is $o(n)$. Finally, two independent exact implementations certify the first finite structural transitions: $n = 15$ is the first non-prefix optimum, while $n = 41$ is the first case of minimum prefix defect greater than one.

Keywords. Pascal's triangle; binomial coefficients; least common multiple; p -adic valuation; Kummer's theorem; Lucas' theorem; extremal number theory.

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1 Introduction and main results

The least common multiple of a complete row of Pascal's triangle is strikingly simple. If

$$\Lambda_r = \text{lcm}_{0 \leq k \leq r} \binom{r}{k},$$

then Farhi [2] proved

$$\Lambda_r = \frac{\text{lcm}(1, 2, \dots, r+1)}{r+1}. \quad (1.1)$$

The complete-row problem goes back at least to Problem E2686 of Montgomery and Breusch [7]; Hong [5] related (1.1) to an identity of Nair, and Guo [4] proved a q -analogue. A separate classical line, beginning with Kummer and Lucas and including Fine [3] and Rowland [9], describes the prime and prime-power divisibility pattern inside a row.

We ask what happens when neither the row nor the selected entries are fixed. For

$$\mathcal{P}_r = \left\{ \binom{r}{0}, \binom{r}{1}, \dots, \binom{r}{\lfloor r/2 \rfloor} \right\}, \quad (1.2)$$

the set of distinct entries of row r , define

$$a(n) = \min_{\substack{r \geq 0, S \subseteq \mathcal{P}_r \\ |S|=n}} \text{lcm}(S). \quad (1.3)$$

Equivalently, $a(n)$ is the smallest positive integer having at least n distinct divisors that occur together in one Pascal row. This is an extremal LCM problem in the spirit of work on subsets of ordinary integer intervals [1], but here the available integers have the rigid arithmetic structure of a Pascal row and the row itself is part of the optimization. We are not aware of a previous study of (1.3) or an equivalent formulation.

The first terms are

$$1, 2, 10, 56, 252, 2310, 12012, 24024, 680680, 1007760, \dots$$

A draft entry for this sequence has been submitted to the OEIS as A399069.

The first nontrivial case already shows why the row cannot be fixed in advance. Row 4 is the first row with three distinct entries,

$$\mathcal{P}_4 = \{1, 4, 6\}, \quad \text{lcm}(\mathcal{P}_4) = 12,$$

but row 5 gives

$$\mathcal{P}_5 = \{1, 5, 10\}, \quad a(3) = 10.$$

Thus moving to a later row can lower the LCM by improving divisibility relations.

Two further phenomena drive the paper. First, although an optimal support may omit strategically expensive coefficients, the complete-row upper bound turns out to have the correct exponential scale. Second, the row in which the optimum occurs is asymptotically forced back toward the first admissible row. The bridge between these global statements and the finite combinatorics is an exact prime-power exclusion principle.

For $1 \leq n \leq |\mathcal{P}_r|$, write

$$A_r(n) = \min_{\substack{S \subseteq \mathcal{P}_r \\ |S|=n}} \text{lcm}(S).$$

For a prime p , put

$$E_p(r) = \max_{0 \leq k \leq \lfloor r/2 \rfloor} v_p \binom{r}{k}, \quad H_{p,e}(r) = \left\{ 0 \leq k \leq \left\lfloor \frac{r}{2} \right\rfloor : v_p \binom{r}{k} > e \right\}.$$

Theorem 1.1 (Prime-power exclusion principle). *Let $d_r = \lfloor r/2 \rfloor + 1$. Then*

$$A_r(n) = \min_{\substack{0 \leq e_p \leq E_p(r) \\ |\cup_{p \leq r} H_{p,e_p}(r)| \leq d_r - n}} \prod_{p \leq r} p^{e_p}. \quad (1.4)$$

Moreover,

$$v_p\binom{r}{k} = \sum_{\nu \geq 1} \mathbf{1}_{\{k \bmod p^\nu > r \bmod p^\nu\}}. \quad (1.5)$$

The theorem turns the choice of binomial coefficients into a weighted exclusion problem: lowering the permitted exponent of a prime power saves multiplicative cost but deletes a precisely determined set of indices. Section 4 develops this point and uses $n = 15$ as a guiding example.

Our first global theorem shows that, despite this freedom, the leading growth is universal.

Theorem 1.2 (Asymptotic growth). *Put*

$$\Phi(x) = \frac{(\log x)^{3/5}}{(\log \log x)^{1/5}}.$$

There exists an absolute constant $c > 0$ such that

$$\log a(n) = 2n + O\left(ne^{-c\Phi(n)}\right). \quad (1.6)$$

Consequently

$$\log a(n) \sim 2n, \quad a(n)^{1/n} \rightarrow e^2.$$

The upper bound comes immediately from the complete row $2n - 2$ and (1.1). The lower bound is the main point. For primes $p > \sqrt{r}$, Lucas' theorem identifies intervals of primes that must divide every admissible LCM. In the j th large-prime band the forced interval has exact length

$$\frac{2n - j - 3}{j(j+1)}.$$

These lengths telescope to $2n$; the Korobov–Vinogradov prime number theorem supplies the quantitative error term.

The same prime-band geometry controls not only the value of the optimum, but also where it occurs. In this sense the optimization is asymptotically rigid: an optimal row is forced back to the first admissible scale, and an optimal support can omit only a sublinear number of positions. Thus Theorem 1.2 determines the *size* of the optimum, while the next theorem determines its *location*. If r_n is an optimal row and $\delta(n)$ is the minimum prefix defect of an optimal support, we obtain the following second main theorem.

Theorem 1.3 (Localization of optimal rows and supports). *There exists an absolute constant $c > 0$ such that every optimal row r_n satisfies*

$$r_n = 2n + O\left(ne^{-c\Phi(n)}\right).$$

Moreover,

$$\delta(n) = O\left(ne^{-c\Phi(n)}\right) = o(n).$$

In particular, $r_n/n \rightarrow 2$.

The final part of the paper returns to exact finite structure. For a support $S \subseteq \{0, \dots, \lfloor r/2 \rfloor\}$, define

$$\delta(S) = 1 + \max S - |S|,$$

the number of missing indices between 0 and $\max S$; thus a prefix has defect 0. Let $\delta(n)$ be the minimum defect among supports attaining $a(n)$.

Exact certification reveals the first two finite transitions. Prefixes are optimal through $n = 14$; at $n = 15$ the minimum defect becomes 1, and among $n \leq 40$ this happens only for $n = 15, 23, 24, 36$. At $n = 41$ the minimum defect jumps to 3. These finite statements are the only computer-assisted results in the paper. They are certified by two independent exact implementations and complete row-by-row output; the analytic results above are independent of those computations.

The paper proceeds from local p -adic structure to global optimization and finally back to finite support geometry. Concretely, Section 3 gives the finite reduction, Section 4 develops prime-power exclusion, Section 5 proves asymptotic growth and localization, and Sections 6–7 provide exact finite certification and independent reproducibility.

2 Preliminaries

2.1 p -adic valuations

For a prime p , let $v_p(m)$ be the exponent of p in m . Legendre's formula gives

$$v_p(m!) = \sum_{\nu \geq 1} \left\lfloor \frac{m}{p^\nu} \right\rfloor, \quad (2.1)$$

and hence

$$v_p \binom{r}{k} = \sum_{\nu \geq 1} \left(\left\lfloor \frac{r}{p^\nu} \right\rfloor - \left\lfloor \frac{k}{p^\nu} \right\rfloor - \left\lfloor \frac{r-k}{p^\nu} \right\rfloor \right). \quad (2.2)$$

We use Kummer's and Lucas' theorems in their standard forms.

2.2 Distinct entries in a row

By symmetry, $\binom{r}{k} = \binom{r}{r-k}$, and

$$\frac{\binom{r}{k+1}}{\binom{r}{k}} = \frac{r-k}{k+1} > 1$$

for $0 \leq k < \lfloor r/2 \rfloor$. Thus $\mathcal{P}_r$ in (1.2) is exactly the set of distinct entries and

$$d_r := |\mathcal{P}_r| = \left\lfloor \frac{r}{2} \right\rfloor + 1. \quad (2.3)$$

In particular, a row can supply n distinct entries only if

$$r \geq 2n - 2. \quad (2.4)$$

2.3 The full-row least common multiple

Let

$$\Lambda_r = \text{lcm}_{0 \leq k \leq r} \binom{r}{k}.$$

By (1.1),

$$\Lambda_r = \frac{\text{lcm}(1, 2, \dots, r+1)}{r+1}. \quad (2.5)$$

Taking p -adic valuations gives

Proposition 2.1. *For every prime p ,*

$$E_p(r) = \lfloor \log_p(r+1) \rfloor - v_p(r+1).$$

2.4 Chebyshev functions

Let

$$\theta(x) = \sum_{p \leq x} \log p, \quad \psi(x) = \sum_{p^\nu \leq x} \log p.$$

Then

$$\psi(x) = \log \text{lcm}(1, 2, \dots, \lfloor x \rfloor).$$

We use the classical Korobov–Vinogradov zero-free-region consequence

$$\theta(x) = x + O\left(xe^{-c_0\Phi(x)}\right), \quad \psi(x) = x + O\left(xe^{-c_0\Phi(x)}\right) \quad (2.6)$$

for some absolute $c_0 > 0$; see, for example, [8, Ch. 6].

3 Basic bounds and finite reduction

Recall

$$A_r(n) = \min_{\substack{S \subseteq \mathcal{P}_r \\ |S|=n}} \text{lcm}(S), \quad a(n) = \min_{r \geq 2n-2} A_r(n).$$

The first observation connects this formulation directly with divisors.

Proposition 3.1. *$A_r(n)$ is the smallest positive integer divisible by at least n distinct elements of $\mathcal{P}_r$. Consequently, $a(n)$ is the smallest positive integer having at least n distinct divisors that occur together in one Pascal row.*

Proof. If $S \subseteq \mathcal{P}_r$, every element of S divides $\text{lcm}(S)$. Conversely, if m is divisible by n distinct elements $x_1, \dots, x_n \in \mathcal{P}_r$, then $\text{lcm}(x_1, \dots, x_n) \mid m$. $\square$

Lemma 3.2. *Every n -element subset of $\mathcal{P}_r$ contains an element at least $\binom{r}{n-1}$. Hence*

$$A_r(n) \geq \binom{r}{n-1}. \quad (3.1)$$

Proof. The half-row entries are strictly increasing, so the n th smallest is $\binom{r}{n-1}$. The LCM is at least the largest selected element. $\square$

Taking $r = 2n - 2$ and using (2.5) gives the two elementary bounds

$$\boxed{\binom{2n-2}{n-1} \leq a(n) \leq \frac{\text{lcm}(1, 2, \dots, 2n-1)}{2n-1}}. \quad (3.2)$$

The sequence $a(n)$ is nondecreasing, although equality can occur.

The same lower bound makes the global minimization finite.

Theorem 3.3 (Finite row reduction). *Suppose $U \geq a(n)$. If*

$$\binom{r}{n-1} > U,$$

then row r cannot attain $a(n)$. In particular, with

$$U_n = \frac{\text{lcm}(1, \dots, 2n-1)}{2n-1}$$

and

$$R(n) = \max \left\{ r \geq 2n - 2 : \binom{r}{n-1} \leq U_n \right\},$$

one has

$$a(n) = \min_{2n-2 \leq r \leq R(n)} A_r(n).$$

Proof. By Lemma 3.2, $A_r(n) \geq \binom{r}{n-1}$. The special choice $U = U_n$ is admissible by (3.2). $\square$

Theorem 3.3 is used twice: asymptotically, to constrain the location of an optimal row, and computationally, to turn every finite claim into an exact finite certificate. The certification algorithm itself is postponed to Section 6, where it is needed.

4 Prime-power exclusion

We now prove Theorem 1.1.

Lemma 4.1. *For every prime p and $0 \leq k \leq r$,*

$$v_p \binom{r}{k} = \sum_{\nu \geq 1} \mathbf{1}_{\{k \bmod p^\nu > r \bmod p^\nu\}}.$$

Proof. Fix $q = p^\nu$, and write $r = aq + s$ and $k = bq + t$ with $0 \leq s, t < q$. The corresponding summand of (2.2) is

$$a - b - \left\lfloor \frac{r - k}{q} \right\rfloor.$$

If $t \leq s$, then $r - k = (a - b)q + (s - t)$ and the summand is 0. If $t > s$, then $r - k = (a - b - 1)q + (q + s - t)$ and the summand is 1. Thus the summand is exactly $\mathbf{1}_{\{t > s\}}$. Summing over ν gives the formula. $\square$

Proof of Theorem 1.1. For exponents $0 \leq e_p \leq E_p(r)$, put $M = \prod_{p \leq r} p^{e_p}$. Then

$$\binom{r}{k} \mid M \iff v_p \binom{r}{k} \leq e_p \text{ for every } p \leq r \iff k \notin \bigcup_{p \leq r} H_{p, e_p}(r).$$

Hence the number of distinct half-row coefficients dividing M is exactly

$$d_r - \left| \bigcup_{p \leq r} H_{p, e_p}(r) \right|.$$

Every admissible exponent vector therefore gives an integer divisible by at least n distinct entries.

Conversely, if S is an n -element support and $L = \text{lcm}\{\binom{r}{k} : k \in S\}$, then $e_p = v_p(L)$ gives an admissible exponent vector, none of the indices in S lies in the union of the corresponding exclusion sets, and $L = \prod p^{e_p}$. The two minima coincide. $\square$

Remark 4.2. Theorem 1.1 separates multiplicative cost from combinatorial loss. Lowering the allowed exponent of p removes precisely the indices in $H_{p, e}(r)$; several such choices combine by taking a union. This is the mechanism behind the holes in optimal supports.

Example 4.3 (The first hole). *Row 34 has 18 distinct entries. Its full-row LCM is Λ_{34} , but lowering the 3-adic budget from 3 to 2 excludes*

$$H_{3,2}(34) = \{8, 17\},$$

while removing the factor 19 excludes

$$H_{19,0}(34) = \{16, 17\}.$$

The union has only three indices,

$$\{8, 16, 17\},$$

so the remaining 15 coefficients all divide

$$\frac{\Lambda_{34}}{3 \cdot 19} = 72\,382\,733\,280.$$

Equivalently, the surviving support is

$$\{0, 1, \dots, 7, 9, 10, \dots, 15\}.$$

Section 6 will certify that this candidate is globally optimal and that $n = 15$ is the first value for which no prefix is optimal. The present example isolates the arithmetic mechanism: one strategically omitted coefficient allows an expensive prime-power requirement to disappear.

5 Asymptotic growth

We now prove Theorem 1.2. The lower bound proceeds in two stages. First, we identify primes that are individually forced into every admissible LCM. Later, in the localization argument, we use the geometry of their exclusion sets to show that even many primes that are not individually forced cannot be omitted simultaneously. The first stage determines the size of $a(n)$; the second determines where an optimum can occur.

5.1 A coarse row bound

Lemma 5.1. *For all sufficiently large n , every row r satisfying $A_r(n) = a(n)$ satisfies*

$$2n - 2 \leq r < 4n.$$

Proof. The lower bound is (2.4). For the upper bound, (3.2) and (2.6) give

$$\log a(n) \leq \psi(2n - 1) - \log(2n - 1) = 2n + o(n).$$

On the other hand, if $r \geq 4n$, then by Lemma 3.2,

$$A_r(n) \geq \binom{4n}{n-1}.$$

Stirling's formula yields

$$\log \binom{4n}{n-1} = (4 \log 4 - 3 \log 3)n + O(\log n),$$

and $4 \log 4 - 3 \log 3 = 2.24934\dots > 2$, a contradiction for large n . □

5.2 Forced primes

Lemma 5.2 (Forced-prime count). *Let $p > \sqrt{r}$ be an odd prime and write*

$$r = jp + s, \quad 0 \leq s < p.$$

Then the number of distinct entries of row r that are not divisible by p is

$$\left\lceil \frac{(j+1)(s+1)}{2} \right\rceil. \quad (5.1)$$

Consequently, p divides the LCM of every n -element subset of $\mathcal{P}_r$ if and only if

$$(j+1)(s+1) \leq 2n-2. \quad (5.2)$$

Proof. Since $p > \sqrt{r}$, one has $r < p^2$ and therefore $j < p$. Thus $r = jp + s$ is exactly the base- p expansion of r with digits (j, s) . Write $k = up + v$, $0 \leq u, v < p$. Lucas' theorem gives

$$p \nmid \binom{r}{k} \iff u \leq j \text{ and } v \leq s.$$

Hence there are exactly

$$T = (j+1)(s+1)$$

p -free indices in the full row $0 \leq k \leq r$.

We now pass from indices in the full row to distinct entries in the half-row. The involution $k \mapsto r - k$ preserves the set of p -free indices because $\binom{r}{k} = \binom{r}{r-k}$.

If r is odd, the involution has no fixed point. Since p is odd, the parity of $r = jp + s$ is the parity of $j + s$; hence for odd r the integers j and s have opposite parity. Thus $T = (j+1)(s+1)$ is even, and exactly $T/2$ p -free indices lie in the half-row.

Now suppose r is even. Then j and s have the same parity, and the only fixed point of the involution is the central index $k = r/2$.

If j and s are both even, then

$$\frac{r}{2} = \frac{j}{2}p + \frac{s}{2}.$$

Both base- p digits of $r/2$ are at most the corresponding digits j, s of r , so Lucas' theorem shows that $\binom{r}{r/2}$ is p -free. In this case T is odd, and the half-row contains $(T+1)/2 = \lceil T/2 \rceil$ p -free entries.

If j and s are both odd, then

$$\frac{r}{2} = \frac{j-1}{2}p + \frac{p+s}{2}.$$

Because $s < p$ and both are odd,

$$\frac{p+s}{2} > s.$$

The lower base- p digit of $r/2$ therefore exceeds the lower digit s of r , so Lucas' theorem shows that the central coefficient is divisible by p . Here T is even, all p -free indices occur in two-element pairs, and exactly $T/2 = \lceil T/2 \rceil$ lie in the half-row.

This proves (5.1) in all cases. The prime p is forced into every LCM of n distinct entries precisely when fewer than n distinct entries are p -free. Since T is an integer,

$$\left\lceil \frac{T}{2} \right\rceil < n \iff T \leq 2n-2,$$

which is (5.2). □

Corollary 5.3 (Forced intervals). *Let $r \geq 2n - 2$ and $j \geq 1$. Put*

$$U_j = \frac{r}{j}, \quad L_j = \frac{r+1}{j} - \frac{2n-2}{j(j+1)}.$$

If $p > \sqrt{r}$ is prime and $L_j \leq p \leq U_j$, then p divides every admissible n -element LCM in row r . Moreover

$$L_j > \frac{r}{j+1} \tag{5.3}$$

and

$$\boxed{U_j - L_j = \frac{2n-j-3}{j(j+1)}}. \tag{5.4}$$

Proof. Within $r/(j+1) < p \leq r/j$ one has $j = \lfloor r/p \rfloor$ and $s = r - jp$. Condition (5.2) becomes

$$(j+1)(r - jp + 1) \leq 2n - 2,$$

which is equivalent to $p \geq L_j$. Also

$$L_j - \frac{r}{j+1} = \frac{r+j+3-2n}{j(j+1)} \geq \frac{j+1}{j(j+1)} > 0$$

by $r \geq 2n - 2$. Finally, direct subtraction gives (5.4). $\square$

The constant 2 in Theorem 1.2 is not an artifact of the complete-row upper bound; it is already encoded in the forced-prime geometry. Ignoring lower-order terms, the j th band contributes a forced interval of length

$$\frac{2n}{j(j+1)},$$

and

$$\sum_{j \geq 1} \frac{1}{j(j+1)} = 1.$$

In other words, the prime bands partition an asymptotic logarithmic mass of $2n$: each band contributes its share, and the shares telescope to the full main term. The proof below makes this principle uniform while allowing the number of bands to grow with n .

Example 5.4 (The forced bands in row 34). *The running example $r = 34$, $n = 15$ also makes the band mechanism concrete. For $j = 1$,*

$$L_1 = 35 - \frac{28}{2} = 21, \quad U_1 = 34,$$

so the first band forces the primes

$$23, 29, 31.$$

For $j = 2$,

$$L_2 = \frac{35}{2} - \frac{28}{6} = \frac{77}{6}, \quad U_2 = 17,$$

so the second band forces

$$13, 17.$$

Thus the same row that exhibits the first optimal hole in Example 4.3 simultaneously illustrates the two mechanisms of the paper: Section 4 explains how selected prime powers can be omitted by sacrificing a few coefficients, while the present section shows that whole intervals of larger primes remain unavoidable.

The dark intervals in Figure 1 drive Theorem 1.2; the incompatibility of the two light intervals drives Theorem 1.3.

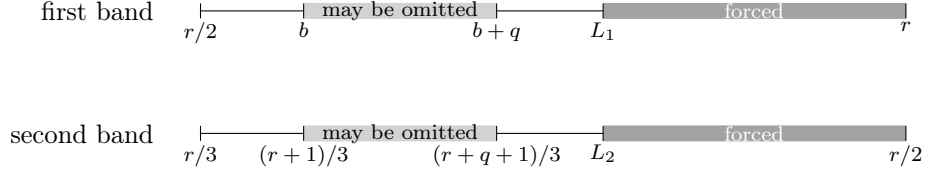


Figure 1: The first two large-prime bands (schematic, not to scale). Dark intervals are individually forced by Corollary 5.3. Light intervals contain primes that a support may omit only when the corresponding exclusion sets fit inside its q omitted positions. The two light masses cannot be removed simultaneously; this is the source of the localization gain.

5.3 Proof of the asymptotic theorem

Proof of Theorem 1.2. For the upper bound, row $2n - 2$ and (2.5) give

$$a(n) \leq \frac{\text{lcm}(1, \dots, 2n - 1)}{2n - 1},$$

so by (2.6),

$$\log a(n) \leq 2n + O\left(ne^{-c_1\Phi(n)}\right). \quad (5.5)$$

For the lower bound, let r be an optimal row. By Lemma 5.1, $2n - 2 \leq r < 4n$ for sufficiently large n . Choose a sufficiently small fixed $\eta > 0$ and set

$$J = \left\lfloor e^{\eta\Phi(n)} \right\rfloor.$$

Then $J = n^{o(1)}$, hence $J < \sqrt{r} - 1$ for large n . For $1 \leq j \leq J$, let $I_j = (L_j, U_j]$. By (5.3),

$$L_j > \frac{r}{j+1} \geq \frac{r}{J+1} > \sqrt{r}$$

for large n . Every prime in I_j is therefore covered by Lemma 5.2 and divides $a(n)$. The intervals lie in disjoint bands, so

$$\log a(n) \geq \sum_{j=1}^J (\theta(U_j) - \theta(L_j)). \quad (5.6)$$

By (5.4),

$$\sum_{j=1}^J (U_j - L_j) = \sum_{j=1}^J \frac{2n - j - 3}{j(j+1)} = (2n - 2) \frac{J}{J+1} - H_J,$$

where H_J is the J th harmonic number. Thus

$$\sum_{j=1}^J (U_j - L_j) = 2n + O\left(\frac{n}{J} + \log J\right). \quad (5.7)$$

It remains to justify the prime-number-theorem error uniformly. Every endpoint occurring in (5.6) lies between

$$X_n := \frac{2n - 2}{J + 1}$$

and $4n$. Since $\log J = O(\Phi(n)) = o(\log n)$,

$$\log X_n = \log n - O(\Phi(n)) = (1 - o(1)) \log n.$$

Consequently

$$\Phi(x) = (1 + o(1))\Phi(n)$$

uniformly for $X_n \leq x \leq 4n$. After reducing the positive constant if necessary, (2.6) therefore gives uniformly at all endpoints

$$\theta(x) = x + O\left(xe^{-c_2\Phi(n)}\right).$$

Since $L_j < U_j = r/j$ and $r < 4n$,

$$\sum_{j=1}^J (U_j + L_j) \leq 2rH_J = O(n \log J).$$

Thus the accumulated error in (5.6) is

$$O\left(n \log J e^{-c_2\Phi(n)}\right) = O\left(ne^{-c_3\Phi(n)}\right),$$

because $\log J = O(\Phi(n))$ can be absorbed by decreasing the exponential constant. Moreover

$$\frac{n}{J} = O\left(ne^{-\eta\Phi(n)}\right),$$

and $\log J$ is negligible on the same scale. Therefore

$$\log a(n) \geq 2n - O\left(ne^{-c_4\Phi(n)}\right). \quad (5.8)$$

Combining (5.5) and (5.8) proves the theorem. $\square$

5.4 A two-band refinement and localization of optimal rows

We now use information that was deliberately discarded in the proof of Theorem 1.2: primes that are not individually forced can still fail to be omitted simultaneously, because their exclusion sets must all fit inside the same set of omitted indices.

Let r satisfy $2n - 2 \leq r < 4n$, put

$$m = \left\lfloor \frac{r}{2} \right\rfloor, \quad q = m + 1 - n, \quad b = \left\lceil \frac{r}{2} \right\rceil,$$

and let $S \subseteq \{0, 1, \dots, m\}$ have $|S| = n$. Write

$$Q = \{0, 1, \dots, m\} \setminus S, \quad |Q| = q.$$

For a prime p , let

$$H_p = \left\{ 0 \leq k \leq m : p \mid \binom{r}{k} \right\}.$$

Then p is absent from the LCM associated with S if and only if $H_p \subseteq Q$.

We shall use the following elementary observation repeatedly.

Lemma 5.5 (Nested exclusion principle). *Let T be a totally ordered finite index set and let $(H_t)_{t \in T}$ be sets satisfying*

$$s \leq t \implies H_s \subseteq H_t.$$

For a fixed set Q , put

$$\mathcal{O} = \{t \in T : H_t \subseteq Q\}.$$

Then $\mathcal{O}$ is an initial segment of T . In particular, if $\mathcal{O} \neq \emptyset$ and $t_0 = \max \mathcal{O}$, then

$$H_t \subseteq Q \iff t \leq t_0,$$

and

$$\bigcup_{t \in \mathcal{O}} H_t = H_{t_0}.$$

Proof. If $t \in \mathcal{O}$ and $s \leq t$, then $H_s \subseteq H_t \subseteq Q$, so $s \in \mathcal{O}$. Thus $\mathcal{O}$ is an initial segment. The remaining assertions are immediate from nestedness. $\square$

The first two prime bands contain particularly simple nested families of exclusion sets. The next lemma records their exact geometry. Since it is used only in the asymptotic argument, we assume $n \geq 4$; then every prime in the second band below is odd.

Lemma 5.6 (Geometry of the first two bands). *Assume $n \geq 4$ and $2n - 2 \leq r < 4n$. Let S, Q, m, q, b be as above.*

In the first band, consider primes

$$b < p \leq b + q.$$

For such a prime,

$$H_p = \{r - p + 1, \dots, m\}, \quad |H_p| = p - b.$$

Let h_1 be the largest value of $|H_p|$ among primes in this interval that are absent from the LCM of S , and put $h_1 = 0$ if there is no such prime.

In the second band, consider primes

$$\frac{r+1}{3} < p \leq \frac{r+q+1}{3}.$$

For such a prime,

$$H_p = \{r - 2p + 1, \dots, p - 1\}, \quad |H_p| = 3p - r - 1.$$

Let h_2 be the largest value of $|H_p|$ among primes in this interval that are absent from the LCM of S , and put $h_2 = 0$ if there is no such prime.

Then

$$\boxed{h_1 + \frac{h_2}{3} \leq q + \frac{(2q - n)_+}{6}}, \tag{5.9}$$

where $(x)_+ = \max(x, 0)$.

Proof. We first determine the exclusion sets exactly from Lucas' theorem.

For a prime in the first band we have $p > b \geq r/2$, hence $r = p + s$ with $0 \leq s < p$ and $m < p$. Thus the base- p digits of r are $(1, s)$, while every $0 \leq k \leq m$ has a single base- p digit k . Lucas' theorem gives

$$p \nmid \binom{r}{k} \iff k \leq s.$$

Therefore

$$H_p = \{s + 1, \dots, m\} = \{r - p + 1, \dots, m\}.$$

Since $b = r - m$,

$$|H_p| = m - (r - p + 1) + 1 = p - b.$$

As p increases, the left endpoint $r - p + 1$ decreases, so these H_p form an increasing nested family. Lemma 5.5 therefore implies that the omitted primes form an initial segment of the prime interval. If $h_1 > 0$, the union of their exclusion sets is the largest omitted set

$$A = \{m - h_1 + 1, \dots, m\} \subseteq Q;$$

for $h_1 = 0$ take $A = \emptyset$.

Now consider a prime in the second band. Since $n \geq 4$ and $r \geq 2n - 2$, such a prime satisfies $p \geq 3$. Moreover,

$$\frac{r}{3} < p \leq \frac{r + q + 1}{3} \leq \frac{r}{2},$$

where the last inequality follows from $q \leq m - 1$. Hence

$$r = 2p + s, \quad 0 \leq s < p,$$

is the base- p expansion of r , with digits $(2, s)$. Also

$$m = \left\lfloor \frac{2p + s}{2} \right\rfloor = p + \left\lfloor \frac{s}{2} \right\rfloor < 2p.$$

Thus every $0 \leq k \leq m$ has one of the following two base- p forms.

If $0 \leq k < p$, then k has digits $(0, k)$. By Lucas' theorem,

$$p \nmid \binom{r}{k} \iff 0 \leq 2 \text{ and } k \leq s,$$

so among these indices divisibility by p occurs exactly for

$$s + 1 \leq k \leq p - 1.$$

If $p \leq k \leq m$, write $k = p + v$ with

$$0 \leq v \leq \left\lfloor \frac{s}{2} \right\rfloor \leq s.$$

The base- p digits of k are $(1, v)$, and Lucas gives

$$p \nmid \binom{r}{k},$$

because $1 \leq 2$ and $v \leq s$. Hence there are no additional divisible coefficients in this part of the half-row. Consequently

$$H_p = \{s + 1, \dots, p - 1\} = \{r - 2p + 1, \dots, p - 1\},$$

and therefore

$$|H_p| = (p - 1) - (r - 2p + 1) + 1 = 3p - r - 1.$$

As p increases, the left endpoint decreases and the right endpoint increases, so these sets are again increasing and nested. Lemma 5.5 shows that the omitted primes form an initial segment of the second prime interval. If $h_2 > 0$, their union is the largest omitted set

$$B = \left\{ \frac{r - 2h_2 + 1}{3}, \dots, \frac{r + h_2 - 2}{3} \right\} \subseteq Q,$$

where the endpoints are integers because $h_2 = 3p - r - 1$ for the largest omitted prime. For $h_2 = 0$ take $B = \emptyset$.

It remains to exploit the fact that both A and B must fit inside the same q -element set Q .

If A and B are disjoint, then

$$h_1 + h_2 = |A \cup B| \leq q,$$

so (5.9) follows immediately.

Assume now that A and B overlap. Write

$$r = 2m + \varepsilon, \quad \varepsilon \in \{0, 1\},$$

so that

$$m = n + q - 1.$$

Let

$$a = m - h_1 + 1$$

be the left endpoint of A and

$$\ell = \frac{r - 2h_2 + 1}{3}$$

the left endpoint of B .

First suppose $a \leq \ell$. Since A ends at m and B ends at most at m , the overlap together with $a \leq \ell$ gives $B \subseteq A$. The inequality $a \leq \ell$ is equivalent to

$$2h_2 \leq 3h_1 - n - q - 1 + \varepsilon.$$

Using $h_1 \leq q$ and $\varepsilon \leq 1$, we obtain

$$h_1 + \frac{h_2}{3} \leq q + \frac{2q - n}{6} \leq q + \frac{(2q - n)_+}{6}.$$

It remains to consider $\ell < a$. Since the intervals overlap, $A \cup B$ is the interval from ℓ to m . The inclusion $A \cup B \subseteq Q$ therefore gives

$$m - \ell + 1 \leq q,$$

or equivalently $\ell \geq n$. Substituting the value of ℓ yields

$$2h_2 \leq 2q - n - 1 + \varepsilon \leq 2q - n.$$

In particular this case is possible only when $2q \geq n$. Since $h_1 \leq q$,

$$h_1 + \frac{h_2}{3} \leq q + \frac{2q - n}{6} = q + \frac{(2q - n)_+}{6}.$$

This proves the lemma. □

Proposition 5.7 (Refined row-wise lower bound). *There exists an absolute constant $c > 0$ such that, uniformly for $2n - 2 \leq r < 4n$ and*

$$q = \left\lfloor \frac{r}{2} \right\rfloor + 1 - n,$$

one has

$$\boxed{\log A_r(n) \geq 2n + \min\left(\frac{q}{3}, \frac{n}{6}\right) - O\left(ne^{-c\Phi(n)}\right).} \quad (5.10)$$

Proof. Fix an n -element support $S \subseteq \{0, \dots, m\}$ and let $L(S)$ denote the LCM of the corresponding coefficients. We prove the asserted lower bound for $\log L(S)$ uniformly in S .

The forced-prime argument used in the proof of Theorem 1.2 is row-wise: uniformly for every $2n - 2 \leq r < 4n$, the primes in the intervals I_j with $1 \leq j \leq J$ contribute

$$2n - O\left(ne^{-c_1\Phi(n)}\right) \quad (5.11)$$

to $\log L(S)$. We now add primes from the parts of the first two bands immediately below the corresponding forced intervals.

Put

$$b = \left\lceil \frac{r}{2} \right\rceil.$$

For the first band define

$$\mathcal{E}_1 = \{p \text{ prime} : b < p \leq b + q\}.$$

For every $p \in \mathcal{E}_1$,

$$H_p = \{r - p + 1, \dots, m\}, \quad |H_p| = p - b \leq q.$$

Thus these are exactly the nonempty exclusion sets in the first band that are individually small enough to fit inside Q . Let h_1 be as in Lemma 5.6. Because the sets are nested, a prime $p \in \mathcal{E}_1$ is absent from $L(S)$ exactly for the primes in the initial subinterval

$$b < p \leq b + h_1.$$

Hence the retained logarithmic prime mass from $\mathcal{E}_1$ is exactly

$$R_1 = \theta(b + q) - \theta(b + h_1). \quad (5.12)$$

All arguments of θ in (5.12) are comparable with n . The Korobov–Vinogradov estimate therefore gives, uniformly,

$$R_1 = q - h_1 + O\left(ne^{-c_2\Phi(n)}\right). \quad (5.13)$$

For the second band define

$$\mathcal{E}_2 = \left\{p \text{ prime} : \frac{r+1}{3} < p \leq \frac{r+q+1}{3}\right\}.$$

For $n \geq 2$ this interval lies inside $(r/3, r/2]$. For every $p \in \mathcal{E}_2$,

$$H_p = \{r - 2p + 1, \dots, p - 1\}, \quad |H_p| = 3p - r - 1 \leq q.$$

Let h_2 be as in Lemma 5.6. Nestedness now shows that the omitted primes in $\mathcal{E}_2$ are exactly those satisfying

$$\frac{r+1}{3} < p \leq \frac{r+h_2+1}{3}.$$

Consequently the retained logarithmic prime mass from $\mathcal{E}_2$ is exactly

$$R_2 = \theta\left(\frac{r+q+1}{3}\right) - \theta\left(\frac{r+h_2+1}{3}\right). \quad (5.14)$$

Again all endpoints are comparable with n , and hence

$$R_2 = \frac{q - h_2}{3} + O\left(ne^{-c_2\Phi(n)}\right). \quad (5.15)$$

The prime sets counted in (5.12) and (5.14) are disjoint from each other and from the forced-prime intervals used in (5.11). For the first band this is exact: the extra interval ends at

$$b + q = r + 1 - n,$$

whereas the forced interval begins at

$$L_1 = r + 2 - n = b + q + 1.$$

For the second band, writing $r = 2m + \varepsilon$ with $\varepsilon \in \{0, 1\}$, one has

$$L_2 - \frac{r + q + 1}{3} = \frac{\varepsilon + 1}{6} > 0.$$

Thus no prime has been counted twice.

Adding (5.13) and (5.15), the extra retained prime mass is

$$\frac{4q}{3} - \left(h_1 + \frac{h_2}{3}\right) - O\left(ne^{-c_2\Phi(n)}\right).$$

Lemma 5.6 therefore gives the lower bound

$$\frac{4q}{3} - q - \frac{(2q - n)_+}{6} - O\left(ne^{-c_2\Phi(n)}\right).$$

The main term equals

$$\begin{cases} q/3, & 2q \leq n, \\ n/6, & 2q \geq n, \end{cases}$$

that is,

$$\min\left(\frac{q}{3}, \frac{n}{6}\right).$$

Combining this with (5.11) proves (5.10) for every support S , and hence for $A_r(n)$. $\square$

Proof of Theorem 1.3. Let r_n be any optimal row and put

$$q_n = \left\lfloor \frac{r_n}{2} \right\rfloor + 1 - n.$$

By Lemma 5.1, $2n - 2 \leq r_n < 4n$ for large n , so Proposition 5.7 applies. Combining it with the upper bound (5.5) gives

$$\min\left(\frac{q_n}{3}, \frac{n}{6}\right) = O\left(ne^{-c\Phi(n)}\right)$$

for some absolute $c > 0$. Since the right side is $o(n)$, the second alternative is impossible for sufficiently large n . Hence

$$q_n = O\left(ne^{-c\Phi(n)}\right).$$

Because

$$q_n = \left\lfloor \frac{r_n}{2} \right\rfloor + 1 - n,$$

we obtain

$$r_n = 2n + O\left(ne^{-c\Phi(n)}\right).$$

Finally, every n -element support in row r_n has prefix defect at most

$$d_{r_n} - n = q_n.$$

Taking the minimum over optimal supports gives

$$\delta(n) = O\left(ne^{-c\Phi(n)}\right) = o(n).$$

$\square$

6 Computer-certified finite structure

The analytic results above are independent of computation. We now return to the finite support geometry suggested by Example 4.3 and certify the first structural transitions exactly.

For $S \subseteq \{0, \dots, \lfloor r/2 \rfloor\}$, recall

$$\delta(S) = 1 + \max S - |S|.$$

For an n -element support, the condition $\delta(S) \leq t$ is equivalent to $\max S \leq n + t - 1$. Thus defect-restricted minima are obtained simply by restricting the allowed indices.

6.1 Exact certification within a row

For fixed r , write $c_k = \binom{r}{k}$, $0 \leq k < d_r$. For a finite set X of positive integers, let

$$\text{MinDiv}(X) = \{x \in X : \text{no distinct } y \in X \text{ satisfies } y \mid x\}.$$

Set

$$D_{0,0} = \{1\}, \quad D_{0,t} = \emptyset \quad (t > 0),$$

and recursively

$$D_{j+1,t} = \text{MinDiv}(D_{j,t} \cup \{\text{lcm}(L, c_j) : L \in D_{j,t-1}\}), \quad (6.1)$$

with the second set omitted for $t = 0$.

Theorem 6.1 (Exact LCM-state dynamic program). *For $0 \leq t \leq d_r$,*

$$\boxed{A_r(t) = \min D_{d_r,t}},$$

with $A_r(0) = 1$.

Proof. Without pruning, (6.1) is the standard subset recurrence. If $L_1 \mid L_2$, then for every positive integer x ,

$$\text{lcm}(L_1, x) \mid \text{lcm}(L_2, x).$$

Hence a state divisible by another state of the same cardinality can never yield a smaller future LCM and may be discarded. Induction on j proves the result. $\square$

Together with the row cutoff from Theorem 3.3, Theorem 6.1 yields a finite exact certificate for every value used below. Section 7 describes the independent second implementation.

6.2 The first non-prefix optimum

The first global departure from prefix optimality occurs at $n = 15$. Example 4.3 already exhibited the winning support in row 34; the purpose of the certification below is to prove that no earlier n and no competing row does better.

Proposition 6.2 (Computer-certified). *For $1 \leq n \leq 14$, $\delta(n) = 0$. At $n = 15$,*

$$\delta(15) = 1, \quad a(15) = 72\,382\,733\,280,$$

with the optimal support from Example 4.3. The best prefix value is 77 636 318 760, attained in row 29.

Certification. The candidate from Example 4.3 and

$$\binom{43}{14} = 78\,378\,960\,360 > a(15)$$

reduce the unrestricted search to $28 \leq r \leq 42$ by Theorem 3.3. The exact minima in these rows are listed in `row_minima.csv` and are reproduced by both certification programs. The defect-restricted certificates show $\delta(n) = 0$ for $n \leq 14$ and give the stated best prefix value at $n = 15$. $\square$

6.3 The first genuinely multi-hole optimum

The second structural transition occurs at $n = 41$: the minimum defect jumps from the previously observed values 0 and 1 to 3. Thus one missing index no longer suffices, and in fact neither do two.

Proposition 6.3 (Computer-certified). *Among $n \leq 40$, the values with $\delta(n) = 1$ are exactly*

$$15, 23, 24, 36,$$

and all remaining values have defect 0. At $n = 41$,

$$\boxed{\delta(41) = 3.}$$

One optimum is attained in row 90 by

$$S_{41} = \{0, 1, \dots, 43\} \setminus \{22, 29, 30\},$$

with

$$a(41) = 235\,699\,793\,161\,402\,293\,458\,658\,402\,494\,400.$$

Exactly five half-row coefficients fail to divide this integer, at indices

$$22, 29, 30, 44, 45.$$

Moreover,

$$\frac{\Lambda_{90}}{a(41)} = 23 \cdot 31 \cdot 47,$$

and the corresponding exclusion sets are

$$H_{23,0}(90) = \{22, 45\}, \quad H_{31,0}(90) = \{29, 30\}, \quad H_{47,0}(90) = \{44, 45\}.$$

Among all 41-element supports of defect at most 2, the least possible LCM is

$$766\,024\,327\,774\,557\,453\,740\,639\,808\,106\,800 = \frac{13}{4}a(41),$$

already attained with defect 1 in row 91.

Certification. The displayed candidate satisfies

$$\binom{122}{40} > a(41),$$

so Theorem 3.3 reduces the unrestricted check to $80 \leq r \leq 121$. Exact unrestricted and defect-restricted row minima are contained in the supplementary tables and are reproduced

independently by both implementations. These data prove the claims about $\delta(n)$ through 41 and show that defect 2 does not improve the best defect-1 value at $n = 41$.

The three displayed exclusion sets give the arithmetic explanation for the optimal pattern in row 90: their union is exactly

$$\{22, 29, 30, 44, 45\}.$$

Thus the computation establishes global optimality, while Theorem 1.1 explains the shape of the support. $\square$

The two examples isolate the first two qualitative changes in optimal support geometry:

$$\text{prefix} \longrightarrow \text{one hole} \longrightarrow \text{genuinely multi-hole}.$$

They are not used in the asymptotic arguments of Section 5.

7 Reproducibility and independent verification

The finite assertions in Section 6 are exact computer-certified statements. The supplementary archive contains source code, complete outputs, machine-readable certificates, and SHA-256 checksums.

The *primary certifier* implements Theorem 6.1 using exact integer LCM states with divisibility pruning. It produces the optimal value, row, support, and final cutoff for every $1 \leq n \leq 41$, together with every unrestricted and defect-restricted row minimum used in the proofs.

The *independent verifier* uses a different state space. It implements Theorem 1.1: prime-power choices are represented by excluded-index bitmasks, unions of masks are formed, and prime-power cost is minimized subject to the allowed number of exclusions. It does not use the LCM-state recurrence of Theorem 6.1.

The two implementations agree on all

$$797 + 2391 = 3188$$

row minima entering the certification. All arithmetic is exact; no floating-point comparisons, heuristic search bounds, probabilistic factorization, or unproved absence assumptions are used. The script `run_all.sh` regenerates the certificates from scratch, while `SHA256SUMS.txt` records checksums of the source and output files.

OpenAI ChatGPT was used as an assistive tool during exploratory problem solving and in drafting portions of the verification code. All mathematical statements, proofs, and computational claims were reviewed by the author; the finite claims are additionally checked by the two algorithmically independent exact implementations described above.

This division of labor is intentional: Sections 4–5 are purely theoretical, whereas the finite “first occurrence” statements of Section 6 are isolated, explicitly certified, and independently reproducible.

8 Discussion

The results give three complementary descriptions of the same optimization problem. Theorem 1.1 is local and arithmetic: it identifies exactly which coefficients are lost when a prime-power budget is lowered. Theorem 1.2 is global and analytic: after all possible local savings are allowed, the surviving large primes still contribute logarithmic mass $2n + o(n)$.

Theorem 1.3 adds rigidity: an optimum cannot realize that saving far out in Pascal’s triangle, but must occur at row $2n + o(n)$.

The finite examples then become structural rather than exceptional. The hole at $n = 15$ and the multi-hole pattern at $n = 41$ are concrete manifestations of the same prime-power tradeoff described by Theorem 1.1. Thus the paper moves from local p -adic constraints to global asymptotics and returns, with that structure in hand, to explain the geometry of optimal finite supports.

9 Outlook

Theorems 1.2 and 1.3 determine the first-order scale of both the value and the location of the optimum. Three finer questions remain natural.

First, what is the true size of

$$r_n - (2n - 2)?$$

Theorem 1.3 makes it $o(n)$, but the correct order may be far smaller. Second, how large can the prefix defect $\delta(n)$ become, and how often is it positive? The certified transitions at 15 and 41 show that non-prefix structure is genuine, while the localization theorem proves only that it is asymptotically sparse. Third, what arithmetic mechanism governs plateaus $a(n) = a(n + 1)$ and possible second-order terms in $\log a(n)$?

All three questions ask for information that is invisible in the leading term $2n$. Their resolution would require a finer understanding of how the prime-power exclusion sets overlap near the first admissible rows.

Data and code availability

All source code, exact output logs, machine-readable certificates, and SHA-256 checksums used to certify the finite results of Section 6 are provided in the supplementary reproducibility archive accompanying this submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the development and preparation of this work, the author used OpenAI ChatGPT to assist with exploratory mathematical discussion, code drafting, organization of the manuscript, and language editing. The author reviewed and edited the resulting material, verified the mathematical content and references, and takes full responsibility for the content of the manuscript.

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